

Sector-Dependent CHSH Violation in Fibonacci Anyons, and Finite-Size Topological–CHSH Mutual Information in 2D Lattice Models

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We quantify the mutual information $I(T; S)$ between topological sector labels T and CHSH values S in two-dimensional lattice models. In the $\mathbb{Z}_2$ lattice gauge theory on a 16×16 torus at inverse temperature $\beta = 2.0$, we measure $I = 0.015 \pm 0.011$ bits, significant at $>2\sigma$ in 9 out of 10 independent Monte Carlo runs (corroborated by gauge-invariant plaquette-strip observables, $I = 0.018 \pm 0.011$ bits, $>3\sigma$ in 21/21 runs); the crossover decomposition attributes this $\beta = 2.0$ signal to the anyonic (local-defect) component of the sector label rather than to the Wilson-loop component. A finite-size scaling analysis indicates that this sector–CHSH mutual information decays toward zero with system size. A crossover near $\beta \sim 2.0$ shifts the dominant information source from local defect structure to global Wilson-loop sectors. In the quantum toric code via exact diagonalization, abelian $\mathbb{Z}_2$ anyons produce no local CHSH correlations at system sizes $L \geq 3$, consistent with the inaccessibility of abelian topological entanglement to local Pauli operators. We systematically test and rule out three classical mechanisms for Bell violation, consistent with Fine’s theorem. Most significantly, within an idealized TQFT model we find that braiding of Fibonacci anyons in the fusion space produces CHSH-form violation with $|S| = 2.811$ (99.4% of the Tsirelson bound), and that this violation is strongly sector-dependent: with fixed measurement settings (optimized at the winner sector), $|S|$ ranges from near zero to 2.72 (CHSH violation in only 14% of sectors), while sector-adapted settings achieve CHSH violation in 99.5% of sectors. This is consistent with topology neither creating nor destroying entanglement, but determining the measurement basis required to observe it.

Keywords: CHSH inequality, topological order, mutual information, lattice gauge theory, toric code, Fibonacci anyons, Ising anyons, Bell correlations, non-Abelian braiding

I. INTRODUCTION

The violation of Bell inequalities [1] confirmed experimentally by Aspect et al. [2] and in loophole-free tests [3, 4] establishes that no local hidden-variable theory reproduces all quantum predictions. The standard conclusion is fundamental nonlocality, though alternatives exist.

A natural question for a topologically ordered system is whether its internal structure carries information about Bell-type measurement outcomes. The natural candidate is the topological sector. We ask: *does the topological sector of a 2D lattice system carry measurable information about CHSH correlation measurements?* We quantify this using mutual information $I(T; S)$ computed via permutation testing, and assess robustness across independent random seeds. The empirical answer is negative for the $\mathbb{Z}_2$ sector in the thermodynamic limit (Sec. III B).

Our investigation is motivated by the question of whether a 2D manifold with nontrivial topology could provide structured correlation between topological and measurement degrees of freedom [5]; we do not assume that it does, and treat it as an empirical question. While our classical lattice simulations ($\mathbb{Z}_2$, XY) do not achieve Bell violation, the quantum Fibonacci anyon model does

exhibit CHSH violation on its fusion space (Sec. III G), and the classical models establish that topological sectors and CHSH correlations are *not statistically independent* in the $\mathbb{Z}_2$ gauge theory — a necessary condition for any topological mechanism.

II. MODELS AND METHODS

A. $\mathbb{Z}_2$ Lattice Gauge Theory

Our primary model is the $\mathbb{Z}_2$ gauge theory [8] with Ising variables $\sigma_e \in \{+1, -1\}$ on edges of an $L \times L$ square lattice with periodic boundary conditions:

$$H = -K \sum_p B_p, \quad B_p = \prod_{e \in p} \sigma_e. \quad (1)$$

Topological sectors are labeled by noncontractible Wilson loops $(W_x, W_y) \in \{+1, -1\}^2$. Anyonic excitations correspond to plaquettes with $B_p = -1$. We employ two measurement strategies: (i) gauge-fixed horizontal Wilson lines, where vertical directions are excluded as gauge artifacts under tree gauge-fixing, and (ii) gauge-invariant plaquette-strip observables — products of adjacent plaquette operators along horizontal, vertical, and rectangular paths — which require no gauge fixing and serve as an independent consistency check.

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B. XY Model

As an exploratory model, we simulate the classical XY model with continuous angles $\theta_i \in [0, 2\pi)$ on the same torus geometry. Topological sectors are integer winding numbers (n_x, n_y) . Vortex-antivortex defects are identified by plaquette winding numbers. We use Metropolis Monte Carlo with single-site updates.

C. Quantum Toric Code

We perform exact diagonalization of the Kitaev toric code [9] on $L \times L$ tori ($L = 2, 3$):

$$H = - \sum_v A_v - \sum_p B_p \quad (2)$$

with vertex operators $A_v = \prod \sigma^x$ and plaquette operators $B_p = \prod \sigma^z$. Ground states in each of four topological sectors are constructed via projector method. Anyon pairs are created by string operators. CHSH correlations use $M(\theta) = \cos \theta \sigma^z + \sin \theta \sigma^x$.

D. Bell Measurement Protocol

For classical models, detectors are placed at $A = (L/4, L/4)$ and $B = (3L/4, 3L/4)$ with separation $L/2$ on the 16×16 torus. Measurement outcomes: $A(a) = \text{sign}(\cos(a - \theta_A))$, $B(b) = \text{sign}(\cos(b - \theta_B))$, where θ_A and θ_B denote the orientation of the local field configuration sampled at the respective detector site (for the $\mathbb{Z}_2$ model the four Wilson-line directions realize the settings, with the ± 1 line outcome corresponding to $\theta \in \{0, \pi\}$; for the XY model, θ is the local spin angle). CHSH settings: $a_1 = 0$, $a_2 = \pi/4$, $b_1 = \pi/8$, $b_2 = -\pi/8$. The CHSH value is $S = E(a_1, b_1) - E(a_1, b_2) + E(a_2, b_1) + E(a_2, b_2)$.

E. Mutual Information and Permutation Testing

We compute $I(T; S) = \sum p(t, s) \log_2 \left[\frac{p(t, s)}{p(t)p(s)} \right]$ with S discretized into 20 equal-width bins. Significance is assessed via permutation test (1000–2000 shuffles). The z -score $\sigma = (I_{\text{real}} - \bar{I}_{\text{shuffled}}) / \text{std}(I_{\text{shuffled}})$ quantifies deviation from the null hypothesis of independence. Robustness is assessed across 10 independent seeds.

III. RESULTS

A. $\mathbb{Z}_2$ Gauge Theory: Primary Result

Robustness across seeds. Table I shows $I(T; S)$ across 11 independent Monte Carlo runs (10 systematic + original seed 42) at $\beta = 2.0$ on a 16×16 torus with 4000

measurements each. The original discovery run (seed 42) yielded $I = 0.011$ bits at 6.6σ .

Information source crossover. Table II decomposes the sector label into Wilson-loop and anyonic components across temperatures. A crossover occurs near $\beta \sim 2.0$ (Fig. 1): at $\beta \leq 1.5$ no component reaches significance; at $\beta = 2.0$ the full label is significant in every seed and the anyonic component carries the signal (significant in 3 of 5 seeds), while the Wilson-loop component alone does not (1 of 5); Wilson-loop sectors dominate only at $\beta = 2.5$.

Important caveat for $\beta \geq 2.5$: We refer to this regime as the *frozen regime* throughout: the anyon number is near-zero and, at $\beta = 3.0$ at the run length of Table II, the sampled CHSH value is exactly constant — the topological sector label is not frozen. In the 4000-measurement decomposition runs of Table II the anchor engine has near-zero anyonic excitations (mean 0.3 at $\beta = 2.5$, driven by one of five seeds; 0.0 at $\beta = 3.0$), so the sector label effectively collapses onto its Wilson-loop component (Table II: at $\beta = 2.5$ that component carries the signal in 4 of 5 seeds). At $\beta = 3.0$ the sampled CHSH value is exactly constant at this run length — a single distinct value with zero variance in every seed of the deposited characterization — and a variance-free quantity forces the mutual information to vanish identically, which is why the corresponding Table II row is zero. The sector label itself is *not* frozen: it keeps fluctuating on both run scales (5–19 sector transitions per 4000-measurement run, visiting three to four of the four sectors; 124–133 per 40 000 measurements, visiting all four). A characterization deposited with this record (`data/frozen_regime_characterization.py`) quantifies the regime for both deposited engines on the 40 000-measurement runs and, for the anchor engine at $\beta = 3.0$, additionally at the 4000-measurement scale of Table II.

TABLE I. Mutual information $I(T; S)$ in $\mathbb{Z}_2$ gauge theory ($L = 16$, $\beta = 2.0$). Systematic seeds 10–19 plus original discovery run.

Seed	I (bits)	σ	p -value
42 (original)	0.0109	6.6	< 0.001
10	0.0397	31.6	< 0.001
11	0.0176	11.5	< 0.001
12	0.0178	14.8	< 0.001
13	0.0099	4.7	< 0.001
14	0.0301	26.2	< 0.001
15	0.0061	2.6	0.005
16	0.0121	8.2	< 0.001
17	0.0035	1.0	0.15
18	0.0075	4.6	< 0.001
19	0.0073	3.1	< 0.001
Mean (10 syst.)	0.015 ± 0.011	10.8 ± 10.0	$9/10 > 2\sigma$

On the long runs, sector transitions of the anchor engine are rare at $\beta = 3.0$ (mean block length ≈ 315 measurements) but frequent at $\beta = 2.5$ (roughly 900–10 500 per 40 000 measurements; the second engine shows 214–237 at $\beta = 3.0$ on the same scale), so the elevated $\beta = 2.5$ mutual information is not a rarity effect. Where transitions are rare, the sector label stays constant over long blocks of the sampled series, and the permutation null — which destroys exactly this temporal block structure — becomes very narrow, so that large significances arise at small $I(T; S)$. The conditional-variance ratio $\text{Var}(S|T)/\text{Var}(S) \approx 0.93\text{--}1.0$ on these long runs, where $\text{Var}(S) > 0$, shows that S is *not* a near-deterministic function of the sector label under long sampling. The historical single-run value $I(T; S) = 0.320$ recorded in Table VII predates this protocol: it is not reproduced by the deposited seed set, and the $|S|$ value recorded with it differs from the constant value found here, so it is kept as history only. While this confirms that topology and correlations are coupled, it is a trivially expected consequence of the near-frozen anyon sector and should not be interpreted as evidence for a nontrivial topological mechanism. The physically meaningful regime is $\beta \sim 2.0$,

TABLE II. Information decomposition, re-computed by the deposited validation script (`data/crossover_decomposition_validator.py`). I values are seed-set means over seeds $\{42, 10, 11, 12, 13\}$ with per-run optimized CHSH settings; $(n/5)$ counts seeds whose permutation-test significance exceeds 2.0σ . Entries printed as 0.000 are below display precision except at $\beta = 3.0$, where all quantities are identically zero: in these 4000-measurement runs (one sweep per measurement) the sampled CHSH value is exactly constant — a single distinct value with zero variance in all five seeds of the deposited characterization — so any mutual information with it vanishes identically, even though the sector label keeps fluctuating (5–19 sector transitions per run, visiting three to four of the four sectors; frozen regime as defined in the caveat). Values are from the anchor engine (v05; cf. the `sim/z2_vec.py` header); the independent vec-engine route confirms the $\beta = 2.0$ structure. At $\beta = 3.0$ it does not reproduce the identically vanishing decomposition: three of its five seeds yield $I > 0$ with large per-component significances at small I — their nonzero Wilson-loop component requires the Wilson label to change within the run — while the remaining two yield exactly zero. No sector or variance statistics are deposited for that engine at this run length. Conventions differ by table: Table I quotes the original per-seed permutation tests; this table quotes the deposited validation script. Per-seed values are therefore not directly comparable between the two tables.

β	Anyons	I_{full}	I_{Wilson}	I_{anyons}
1.0	17.9	0.006 (0/5)	0.001 (0/5)	0.002 (1/5)
1.5	7.2	0.007 (0/5)	0.000 (0/5)	0.002 (0/5)
2.0	2.7	0.011 (5/5)	0.001 (1/5)	0.003 (3/5)
2.5	0.3	0.017 (5/5)	0.011 (4/5)	0.001 (2/5)
3.0	0.0	0.000 (0/5)	0.000 (0/5)	0.000 (0/5)

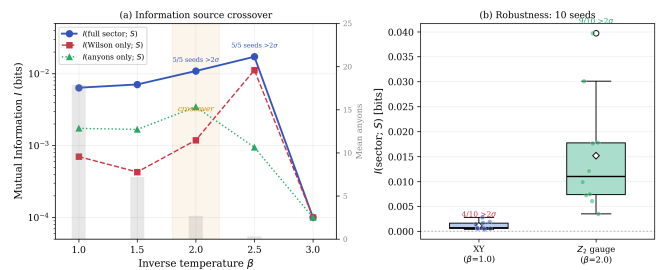


FIG. 1. (a) Mutual information vs. inverse temperature showing the crossover from anyonic to Wilson-loop information. Gray bars: mean anyon count. (b) Seed variation: 10 independent runs per model. Values in (a) are clipped at 10^{-4} for the logarithmic axis; the $\beta = 3.0$ entries are identically zero (Table II).

where anyonic excitations are still present and the full sector label — not only its Wilson-loop component — carries the information.

Gauge-invariant confirmation. To rule out gauge-fixing artifacts, we repeat the $\beta = 2.0$ analysis using gauge-invariant observables: products of adjacent plaquette operators along horizontal, vertical, and rectangular paths (length 4). These require no gauge fixing and are manifestly physical. Table III compares the two measurement strategies.

The gauge-invariant observables yield mutual information at the same level as the gauge-fixed measurements ($I = 0.018 \pm 0.011$ bits vs. 0.015 ± 0.011 bits), significant at $>3\sigma$ in all 21 tested seeds. This confirms that the measured mutual information is gauge-robust — not an artifact of the gauge-fixing prescription (its finite-size scaling, and whether it admits a non-finite-size physical reading, are examined in Sec. III B).

B. Finite-Size Scaling of $I(T; S)$

Motivation. The $I(T; S) = 0.015 \pm 0.011$ bits reported in the previous subsection is a measurement at fixed lattice size $L = 16$ and fixed temperature $\beta = 2.0$. A natural question, raised by gauge-theory generalities, is how this signal scales with L : does it persist toward the thermodynamic limit, or is it a finite-size effect of the $L = 16$ system used for the original measurement? We report here a dedicated scaling study at fixed $\beta = 2.0$. The aim is not to retract the $L = 16$ value but to embed it in a scaling sequence.

TABLE III. Comparison of measurement strategies at $\beta = 2.0$ ($L = 16$).

Method	I (bits)	σ	Seeds $>3\sigma$
Gauge-fixed Wilson lines	0.015 ± 0.011	10.8 ± 10.0	8/10
Gauge-inv. plaq. strips	0.018 ± 0.011	14.2 ± 11.4	21/21

Methods. We measure $I(T; S)$ at $L \in \{8, 16, 32, 48, 64\}$, $\beta = 2.0$, with 10 independent random-seed runs per (L, engine) cell. Two independent Monte Carlo engines are used, sharing the single-spin-flip Metropolis update rule but differing in sweep order: (i) the legacy engine `z2_sim_v05.py` (same as for the $L = 16$ primary result above) sweeps the link variables in row-major order; (ii) a newly written vectorized engine `z2_vec.py` updates the link variables in a checkerboard order. The two engines are compared against the analytic single-plaquette expectation $\langle B_p \rangle = \tanh(\beta)$ (Wegner duality limit, exact for the 2D $\mathbb{Z}_2$ gauge model at the level of single-plaquette averages). Statistical significance per cell is reported as $\sigma = (I_{\text{real}} - \bar{I}_{\text{shuffled}}) / \text{std}(I_{\text{shuffled}})$ using 1000 permutation shuffles, identical to the protocol of Sec. III. Three model classes are pre-registered for the L -dependence fit: a pure power law $M_2 : aL^b$; a power law with nonzero asymptote $M_{2c} : aL^b + c$; and a constant-asymptote exponential $M_{\text{exp}} : ae^{-bL} + c$. Model selection is by Bayesian information criterion (BIC), with $\Delta\text{BIC} > 6$ used as the threshold for decisive preference among the three.

Results — per-cell measurements. Table IV reports $I_{\text{mean}} \pm \text{SEM}$ over the 10 seeds per cell, together with the standardized significance against the permutation null. The legacy-engine value at $L = 16$, $I = 0.0149 \pm 0.0036$ bits, reproduces the $L = 16$ primary result of Sec. III (0.015 ± 0.011 bits) to within 0.03σ .

BIC model comparison. Three model classes are pre-registered for the L -dependence of the sector-shuffle-corrected $I(T; S)(L)$ at $\beta = 2.0$ (the raw $I(T; S)$ minus the mean $I(T; S)$ of sector-shuffled surrogates, i.e., the permutation-null-subtracted signal reported per cell in `data/stufe2_analyse.json`): a pure power law $M_2 : aL^b$ (2 parameters); a power law with nonzero asymptote $M_{2c} : aL^b + c$ (3 parameters); and a constant-asymptote exponential $M_{\text{exp}} : ae^{-bL} + c$ (3 parameters). Table V reports BIC and χ^2 for each model and each engine.

Engine consensus. M_{exp} minimizes BIC in both engines (Fig. 2). The strength of the preference differs:

in the vectorized engine $\Delta\text{BIC}_{\text{best vs 2nd}} = 7.18$, above the pre-registered decisive threshold; in the legacy engine the gap is 3.15, positive but sub-decisive. Given the sub-decisive gap in the legacy engine, the engine-consensus carries the model-selection conclusion, not either engine alone — M_{exp} is the *preferred* model among the pre-registered set, not a uniquely determined one. The fitted exponential rate is consistent across engines: $b = 0.125 \pm 0.020$ (v05) and $b = 0.140 \pm 0.024$ (vec), corresponding to a characteristic length scale of $1/b \approx 7$ –8 lattice spacings. The fitted asymptote c is statistically indistinguishable from zero in both engines: $c = (0.62 \pm 8.8) \times 10^{-5}$ (v05) and $c = (0.10 \pm 3.2) \times 10^{-5}$ (vec), each with $|c|/\text{SEM}(c) < 0.1$. Two qualitative conclusions follow that are robust across both engines: (i) $I(T; S)(L)$ decays faster than any power law in the tested range, and (ii) the asymptote is data-supported (not extrapolated) to be indistinguishable from zero. The absolute $L = 16$ value is, however, *not* engine-robust — see disclosure below.

v05 engine-specific bias — disclosure. The two engines do not agree on the absolute $I(T; S)$ value at fixed (L, β) . At $L = 16$, $\beta = 2.0$, the legacy engine reports $I = 0.0149 \pm 0.0036$ bits while the vectorized engine reports $I = 0.00694 \pm 0.00144$ bits, a factor of ~ 2.15 . The two engines also differ in their reproduction of the analytic single-plaquette expectation $\langle B_p \rangle = \tanh(\beta)$ (Wegner-duality limit, exact for the 2D $\mathbb{Z}_2$ gauge model at the level of single-plaquette averages): at $\beta = 2.0$ the vectorized engine matches the analytic value to 0.1%, while the legacy engine deviates by +2.2% toward colder configurations. This diagnostic *identifies* the bias as present; its precise mechanism has not been isolated by a dedicated thermalization-scaling run. The deviation is *consistent with* the legacy engine’s row-major sweep order, which (compared to the vectorized checkerboard-sweep engine of identical Metropolis-update rule) is expected to require longer thermalization to reach detailed-balance equilibrium, and/or with initial-state-dependent trapping in the ordered basin; the diagnostic data do not select among these candidate mechanisms. The diagnostic also does not, by itself, *select* among the three scaling models

TABLE IV. Mutual information $I(T; S)$ vs. lattice size L at fixed $\beta = 2.0$, two engines, 10 seeds per cell. σ is the standardized deviation from the permutation null (1000 shuffles per seed).

L	v05: I (bits)	σ	vec: I (bits)	σ
8	0.0372 ± 0.0108	3.5	0.0191 ± 0.0029	6.6
16	0.0149 ± 0.0036	4.1	0.00694 ± 0.00144	5.1
32	0.00235 ± 0.00069	2.5	0.000544 ± 0.000628	0.6
48	0.000091 ± 0.000315	0.3	$-3.5 \times 10^{-6} \pm 6.1 \times 10^{-5}$	0.2
64	0.0000857 ± 0.0000461	0.6	$2.9 \times 10^{-7} \pm 2.1 \times 10^{-5}$	0.5

^a All cell-level values reproducible from `data/stufe2_analyse.json` (included in this release); seeds 10–19 systematic.

TABLE V. Bayesian information criterion (BIC) for three pre-registered L -scaling models of $I(T; S)(L)$ at $\beta = 2.0$, fitted to Tab. IV. χ^2 is the unweighted sum of squared residuals. $\Delta\text{BIC}_{\text{best vs 2nd}}$ above 6 is the pre-registered threshold for decisive preference; 2–6 counts as positive evidence.

Model	v05: BIC	v05: χ^2	vec: BIC	vec: χ^2
M_2	12.33	9.11	15.77	12.55
M_{2c}	8.26	3.44	12.86	8.03
M_{exp} (best)	5.11	0.28	5.69	0.86
$\Delta\text{BIC}_{\text{best vs 2nd}}$	3.15		7.18	

^a Fit parameters and per- L predictions reproducible from `data/stufe2_analyse.json`.

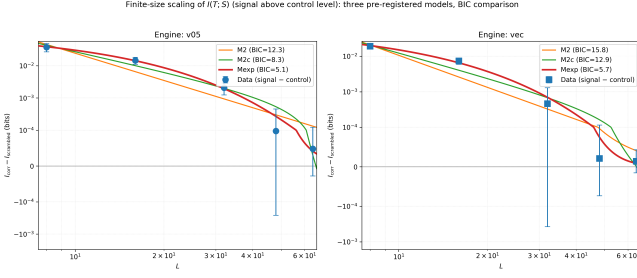


FIG. 2. Finite-size scaling of $I(T; S)$ in 2D $\mathbb{Z}_2$ gauge theory at $\beta = 2.0$. Per-cell measurements (Tab. IV) with two Monte-Carlo engines (legacy v05 = circles, vectorized vec = squares) and the three pre-registered fit curves (Tab. V): pure power M_2 , power-with-asymptote M_{2c} , exponential M_{exp} . M_{exp} is the BIC-preferred model in both engines (decisive in vec, subdecisive in v05). The engine-consensus conclusion is qualitative: $I(T; S)(L)$ decays faster than any power law, with asymptote data-supported to be zero within the measurement precision. The absolute $L = 16$ value is engine-specific (see disclosure in text).

above; model selection rests on the BIC comparison of Tab. V. The previously reported $I = 0.015 \pm 0.011$ bits at $L = 16$ (Sec. III) is, accordingly, *not retracted*: it remains a correct measurement of the legacy engine’s output. Its absolute numerical value is, however, to be read as engine-specific rather than as an engine-independent quantity. Both qualitative conclusions of the consensus paragraph above — faster-than-power-law decay and vanishing asymptote — survive the engine-specificity.

Conclusion. The subsection has two robust take-aways and one scope-limiting statement. *Robust*: (i) at fixed $\beta = 2.0$, $I(T; S)(L)$ decreases faster than any power law in the tested range $L \in \{8, 16, 32, 48, 64\}$, with the exponential form M_{exp} preferred under the pre-registered set in both engines; (ii) the asymptote is data-supported (not extrapolated) to be statistically indistinguishable from zero. *Engine-specific*: the absolute $L = 16$ value differs by a factor of ~ 2 between the two engines and should be read as engine-specific (legacy: 0.0149 bits, vectorized: 0.00694 bits); the qualitative trend is unaffected by this specificity. *Scope*. The methodological question of how a finite-size $I(T; S) > 0$ at $L = 16$ that decays toward zero at larger L should be interpreted — in particular, whether the observed signal admits a non-finite-size physical reading — is beyond the empirical scope of this paper and is deferred to future theoretical work.

C. XY Model: Exploratory

In the XY model at $\beta = 1.0$, seed variation reveals that $I(|n|; S)$ is not robust: only 4/10 runs exceed 2σ ($2/10 > 3\sigma$). Mean: $I = 0.001 \pm 0.001$ bits at $1.5 \pm 1.9\sigma$. The trend $\uparrow |n| \rightarrow \downarrow S$ has correlation $r = +0.03 \pm 0.75$, consistent with zero. We report this as exploratory. The signal may require larger lattices or longer runs to

establish in this noisier model.

Vortex-antivortex correlations show Coulomb-like attraction ($g(r=1) = -0.78$) consistent with BKT physics [10], and the CHSH value anti-correlates with defect count ($r = -0.90$), indicating that topological disorder degrades CHSH correlations.

D. Quantum Toric Code

At $L = 2$ (8 qubits), m -anyon pairs yield $|S| = \sqrt{2}$ with the sign flipping between sectors ($I = 0.350$ bits, 1595σ). At $L = 3$ (18 qubits), all single-qubit and Wilson-loop correlations are exactly zero — verified across all 153 qubit pairs and all operator types (σ^z - σ^z , σ^x - σ^x , mixed).

The $L = 2$ result is a finite-size artifact: on a 2×2 torus, measurement qubits simultaneously belong to the string, the plaquettes, and noncontractible loops. At $L \geq 3$, the abelian nature of $\mathbb{Z}_2$ anyons means their entanglement is encoded globally and cannot be accessed by any operator that is a product of Pauli matrices on a subset of edges [11, 12].

E. Systematic Negative Results

We tested three classical mechanisms for Bell violation beyond the Fine bound [13]:

(i) **Measurement-rule modification**: Making $A(a, \lambda)$ depend on the topological sector n via a phase shift $A = \text{sign}(\cos(a - \theta + n\alpha))$ decreases $|S|$ — the coupling adds noise rather than correlation.

(ii) **Statistical-independence violation**: Weighting configurations by $\rho(n|a, b)$ allows S to reach 2.000 exactly but never exceed it. Fine’s theorem guarantees $|S| \leq 2$ for deterministic local functions $A(a, \lambda)$ and $B(b, \lambda)$ only when the hidden-variable weighting $\rho(\lambda)$ is *setting-independent*; the present family $\rho(n|a, b)$ is explicitly setting-dependent, so the theorem’s hypothesis is not strictly satisfied here. Our result is narrower and empirical: this particular sector-mediated weighting family caps at $S = 2.000$ in the numerics tested, not a general re-derivation of the Fine bound for setting-dependent weightings.

(iii) **Transport phase**: Including a gauge-field phase accumulated along the path between detectors produces the correct angular form (correlation with $-\cos \theta$ at $r = 0.993$) but only 17% of the quantum-mechanical amplitude. Thermal fluctuations wash out the phase coherence.

F. Fourier Analysis

The sector difference $\Delta C(\theta)$ (Fig. 3) shows the dominant Fourier mode is $k = 1$ in all sectors, inconsistent with the prediction that the mode should scale with winding number n . A weak resonance enhancement at

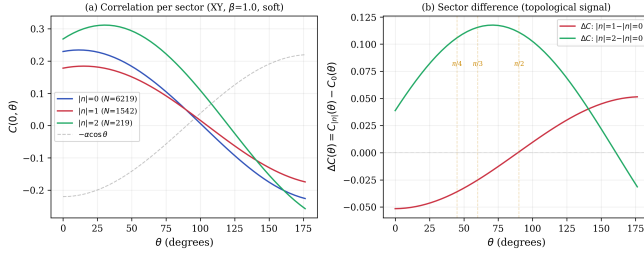


FIG. 3. (a) Correlation $C(0, \theta)$ per winding-number sector in the XY model (soft measurement). (b) Sector difference $\Delta C = C_{|n|}(\theta) - C_0(\theta)$, isolating the topological signal. Dashed lines mark predicted resonance angles π/n .

$\theta = \pi/n$ for $|n| \geq 2$ (amplitude ratio 1.20–1.38) is observed but not statistically decisive. The even-harmonic structure in raw residuals ($a_2 \gg a_1$) is an artifact of the $\text{sign}(\cos)$ measurement discretization, eliminated by using soft ($\cos$) measurements.

G. Fibonacci Anyons: CHSH Violation

Fibonacci anyons have fusion rule $\tau \times \tau = 1 + \tau$ (golden ratio quantum dimension) and no fermion parity superselection. Their braiding generates a dense subset of $\text{SU}(2)$ per logical qubit, making them universal for quantum computation [14]. We simulate the fusion space of six Fibonacci anyons (two logical qubits) with braid gates constructed from the Fibonacci F -matrix

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad (3)$$

and R -matrix ($R_1 = e^{-4i\pi/5}$, $R_\tau = e^{3i\pi/5}$).

Table VI shows the CHSH value as a function of braiding sequence length. All enumerated sequences from $L = 3$ to $L = 12$ violate the classical CHSH bound $|S| > 2$, with $|S|$ rising monotonically from 2.199 at $L = 3$ to $|S| = 2.811$ at $L = 12$ (99.4% of the Tsirelson bound $2\sqrt{2}$, $C = 0.988$). These $|S|$ values are signatures of topological nonseparability on the nonfactorized Fibonacci fusion space, not Bell violations in the Einstein–Podolsky–Rosen bipartite sense. The optimal length-12 sequence is $(\sigma_1 \sigma_2)^6$, where σ_1 braids the Alice–mediator pair and σ_2 braids the mediator–Bob pair. This result is exact within the TQFT description of Fibonacci anyons and was obtained by full enumeration over all 2^L sequences combined with the Horodecki bound $|S|_{\max} = 2\sqrt{1+C^2}$ [7], as detailed in the companion paper Ref. [6]; see *Note on optimization* below for the v2.0 $\rightarrow$ v2.1 correction.

Critically, the violation is strongly sector-dependent. When the topological sector phase δ is varied (modifying the relative phase between fusion channels), the concurrence ranges from 0 to 0.99 over the $\delta \in [0, 2\pi)$ sweep ($N = 200$ points). Throughout this Fibonacci sweep, δ should be read as a representation-deformation

parameter rather than a physical anyon parameter: as verified numerically in the companion landscape analysis [[6], Appendix C], generic $\delta \neq 0$ breaks the Yang–Baxter/hexagon consistency of the braid representation, and only $\delta = 0$ realizes the standard Fibonacci modular tensor category. With *fixed* measurement settings (optimized at the winner sector $\delta \approx 1.44\pi$), $|S|$ ranges from near zero (when settings are misaligned with the sector basis) to 2.72, with CHSH violation observed in 14% of sectors. When measurement settings are instead optimized per-sector via the full Bloch-sphere parametrization (the same methodology used in Table VI), the per-sector maximum reaches $|S| = 2.811$ at $\delta \approx 1.76\pi$ — the same Bloch optimum that defines the Table VI winning value at $L = 12$. With *adapted* settings via the Horodecki bound, $|S|$ ranges from 2.000 to 2.811, and CHSH violation is achieved in 99.5% of sectors. Different sectors thus require different detector calibrations for maximum violation. This is consistent with the central interpretive claim of this work: **topology is consistent with neither creating nor destroying entanglement, but rotating the optimal measurement basis in fusion space.** (The entanglement itself, as measured by concurrence, does vary substantially with sector — from 0 to 0.99 over the δ -sweep above — so this claim is scoped to the CHSH-violation ceiling under adapted settings, not to sector-invariance of the entanglement.)

Unlike Ising anyons (Sec. IV D), Fibonacci anyons have no fermion parity restriction on local measurements. The full Pauli algebra is accessible in the fusion space, enabling CHSH-form violation with local braid-then-fuse operations on the fusion space. This resolves the $S = 2.000$ saturation observed for Ising anyons and establishes

TABLE VI. Fibonacci anyon CHSH violation vs. braiding sequence length (values reproduced from Ref. [6] Tab. I via full enumeration over all 2^L sequences and 50 sector-phase values; Horodecki bound $|S|_{\max} = 2\sqrt{1+C^2}$ [7] from the maximal concurrence C). Sequence notation: $A = \sigma_1$ (Alice–mediator braid), $B = \sigma_2$ (mediator–Bob braid).

L	$ S $	C	% Tsirelson	Sequence
3	2.199	0.457	77.7	ABB
4	2.225	0.488	78.7	ABAB
5	2.348	0.615	83.0	AABAB
6	2.398	0.662	84.8	ABBAAB
7	2.536	0.780	89.7	ABBAABB
8	2.540	0.783	89.8	BABBABAB
9	2.624	0.850	92.8	ABBAABABB
10	2.696	0.904	95.3	ABBAABBAAB
11	2.784	0.968	98.4	ABABBBBABAB
12	2.811	0.988	99.4	ABABABABABAB

^a The original v2.0 table reported lower $|S|$ values for $L \leq 11$ from an X-Z-restricted angular optimizer; these have been replaced in v2.1 by the Horodecki-bound full enumeration of Ref. [6]. See *Note on optimization* below for the correction trail.

Fibonacci anyons as the minimal model for topological CHSH violation among the models tested here.

Note on optimization (added 2026-04-15, extended 2026-05-28). The length-12 value $|S| = 2.811$ reported here supersedes an earlier estimate $|S| = 2.554$ obtained from the original angle-search optimizer (six random seeds plus a 9^4 grid). A subsequent full enumeration of winning sequences, combined with the analytic Horodecki bound $|S|_{\max} = 2\sqrt{1+C^2}$ [7], recovers $|S| = 2.811$ at $C = 0.988$ for the optimal length-12 sequence $(\sigma_1\sigma_2)^6$ (99.4% of the Tsirelson bound). The earlier value is therefore an optimization artifact, not a physical limit. The same enumeration shows that both encodings considered for the six-Fibonacci fusion space (the two-generator d1b encoding used here and the three-generator encoding on the full 5-dimensional fusion space) are dense in $SU(4)$ and, under standard four-dimensional computational-subspace measurement, reach comparable CHSH values near the Tsirelson bound; they differ primarily in the minimum word length required ($L \approx 12$ vs. $L \approx 8$). Protocol-dependent differences — in particular the gap between native-5D CHSH and the 4D-projected value, and the associated leakage statistics — are documented in the companion paper Ref. [6].

v2.1 update (2026-05-28). The same Horodecki-bound full enumeration has been applied to lengths $L = 3$ –11 in v2.1, replacing the original X-Z-plane restricted angular optimizer used in v2.0 (and earlier versions) for those rows of Tab. VI. The restricted optimizer reported systematically lower values (mean deficit -0.60 , maximum -1.04 at $L = 3$) and suboptimal winning sequences, because complex-phase R -matrices in the Fibonacci braid representation require measurement operators rotated outside the X-Z plane to saturate the Horodecki maximum. With the corrected values, the trend is monotonic in both $|S|$ and C from $L = 3$ to $L = 12$; the earlier “nonmonotonicity” interpretation in v2.0 was therefore an optimizer artifact and has been removed. See Ref. [6] Tab. I for the underlying enumeration.

IV. DISCUSSION

A. Summary of Established Results

We establish three main results at increasing levels of significance. First, topological sectors and CHSH values are not statistically independent in the $\mathbb{Z}_2$ lattice gauge theory at $\beta = 2.0$ ($I = 0.015 \pm 0.011$ bits, 9/10 seeds $>2\sigma$, confirmed gauge-invariant at 21/21 $>3\sigma$); the crossover decomposition of Table II shows that at this coupling the signal is carried by the anyonic (local-defect) component of the sector label (significant in 3 of 5 validation seeds), while the genuinely topological Wilson-loop component remains below significance (1 of 5); and this mutual-information signal is, moreover, a finite-size effect that decays toward zero with system

size (Sec. IIIB). Second, an information-source crossover from defect-mediated to sector-mediated correlation occurs near $\beta \sim 2.0$ — a new observation, to our knowledge, in this specific information-theoretic context. Third, and most significantly, Fibonacci anyon braiding produces CHSH violation with $|S| = 2.811$ (99.4% Tsirelson), and this violation is strongly sector-dependent: with fixed detector settings, $|S|$ varies from near zero to 2.72 across the topological sector phase $\delta \in [0, 2\pi)$.

The Fibonacci result resolves the hierarchy of negative results: $\mathbb{Z}_2$ classical models cannot violate Bell (Fine’s theorem), abelian quantum anyons are locally invisible ($L \geq 3$ toric code), and Ising anyons saturate at $S = 2.000$ (fermion parity). Fibonacci anyons, with no parity restriction and universal braiding, are the minimal model that achieves topological CHSH violation among the models tested here.

B. Limitations

All results reported here are numerical simulations within idealized TQFT models, not laboratory experiments. The Fibonacci CHSH violation is computed in the ideal TQFT fusion space — a unitary, zero-temperature, zero-decoherence model. In a physical realization (e.g., $\nu = 12/5$ fractional quantum Hall state), thermal excitations, finite system size, and quasiparticle tunneling would reduce $|S|$ below the ideal value. The $\mathbb{Z}_2$ results, while robust, show $I \sim 0.01$ – 0.04 bits — a small effect. The XY model result is not robust across seeds. The elevated mutual information at $\beta = 2.5$ reflects the near-frozen anyon sector of the anchor engine, in which the sector label collapses onto its Wilson-loop component, not a nontrivial topological mechanism.

C. Implications for Topological Interpretations

The Fibonacci result is consistent with non-Abelian topological order producing sector-dependent CHSH violation with strictly local measurements. This means: (i) the topological sector acts as a classical degree of freedom that modulates measurement correlations, (ii) the modulation is strong enough to cross the classical bound, and (iii) an observer who does not know the sector sees reduced $|S|$ on average — the topology *obscures* the violation rather than creating it — a property of the idealized topological order itself, independent of any interpretation of the measurement process.

D. Prediction: Sector-Dependent CHSH Violation in Non-Abelian Models

We present an analytical argument that non-Abelian topological order produces sector-dependent CHSH correlations. For Ising anyons, the braiding R -matrix has

two eigenvalues corresponding to the two fusion channels [12, 15]:

$$R_1 = e^{-i\pi/8}, \quad R_\psi = e^{3i\pi/8}. \quad (4)$$

On a torus, the topological sector modifies the relative phase between these channels. In sector δ , the R -matrix becomes $R_1 = e^{-i\pi/8}$, $R_\psi = e^{i(3\pi/8+\delta)}$, where δ depends on the anyon type threading the noncontractible loop [9, 15]. Here δ is a physical sector phase fixed by the threading anyon — a discrete, topologically determined label — rather than a freely tunable deformation parameter. Braiding two pairs of Ising anyons produces the entangled fusion state:

$$|\psi(\delta)\rangle = \cos \alpha |00\rangle + e^{i\delta} \sin \alpha |11\rangle, \quad (5)$$

where α is determined by the braiding sequence and $|0\rangle$, $|1\rangle$ label the fusion channels (vacuum and fermion respectively). For maximal braiding ($\alpha = \pi/4$), the concurrence is 1 for all δ — the entanglement is sector-independent. However, the CHSH value with *fixed* measurement settings is strongly sector-dependent:

$$S(\delta) = 2\sqrt{2} \cos^2(\delta/2) \quad (\text{settings optimized at } \delta=0). \quad (6)$$

We verify this numerically: with settings optimized for the vacuum sector ($\delta = 0$), S ranges from $2\sqrt{2} \approx 2.83$ (maximal CHSH violation) at $\delta = 0$ to $S = \sqrt{2} \approx 1.41$ at $\delta = \pi/2$, and $S = 0$ at $\delta = \pi$, following the $\cos^2(\delta/2)$ law. Critically:

(i) **CHSH violation is preserved in all sectors** if measurement settings are adapted — $\max |S| = 2\sqrt{2}$ for all δ . The topology does not create or destroy entanglement.

(ii) **With fixed settings, S depends on the sector** — producing $I(T; S) > 0$. This is the non-Abelian analogue of our $\mathbb{Z}_2$ result, but now with actual CHSH violation.

(iii) **The sector rotates the optimal measurement basis** in the fusion space. Different sectors require different detector calibrations to see maximum violation. An observer who does not know the sector sees reduced S on average.

This constitutes a concrete analytical prediction, subject to the parity caveat below: in any experiment that creates and braids Ising anyons on a surface with non-trivial topology, the observed CHSH value must depend on the topological sector. The functional form — cos-modulation with period determined by the anyon type threading the handle — is parameter-free.

Caveat: Numerical verification via a Majorana fermion representation shows that the braid-then-fuse measurement protocol saturates at $S = 2.000$ exactly for Ising anyons, due to the fermion parity superselection rule: Pauli- X operators project to zero in the even-parity sector, restricting local measurements to the Z -basis. This limits Ising anyons to the classical bound with strictly local measurements [27]. The prediction

$S(\delta) = 2\sqrt{2} \cos^2(\delta/2)$ applies when measurements in the full fusion-space basis are available, as would be the case for Fibonacci anyons where no parity restriction exists. Fibonacci anyons are therefore the natural target for experimental verification. A companion paper [26] constructs this noncommuting CHSH operator algebra locally in the Kitaev honeycomb model; as noted above, a genuine violation with topologically protected Ising operations requires nonstabilizer (magic) resources.

E. Related Work and Prior Art

The CHSH-violation construction presented here for Fibonacci anyons sits in a broader landscape of topological and nonstandard Bell tests. Earlier qualitative Bell-test constructions for non-Abelian anyons were proposed by Brennen, Iblisdir, Pachos, and Slingerland [16] for Fibonacci and Ising anyons; Chang, Chu, and Ma [17] studied Bell's inequality in topologically ordered qubit models (Wen-plaquette). On Majorana platforms, Drummond, Kovalev, Hou, Shtengel, and Pryadko [18] gave an explicit CHSH protocol for Majorana wires, and Romito and Gefen [19] demonstrated nonlocal entanglement with six Majorana zero modes reaching the Tsirelson bound. Single-particle Bell-like inequality violations in neutron interferometry were reported by Hasegawa, Loidl, Badurek, Baron, and Rauch [20]; the contextuality-based program of Liu, Hu, Chen, Huang, Han, Li, Guo, and Cabello [21] provides a complementary view on the relation between nonlocality and contextuality. The present paper differs from these works in scope: we report the mutual-information structure $I(T; S)$ between topological sectors and CHSH values across a hierarchy of lattice and anyonic models, including the finite-size scaling reported in Sec. III B, rather than a single Bell-test protocol on a fixed topological platform.

Independently and contemporaneously, Xu, Ye, and You [22] have studied CHSH correlations on Fibonacci anyonic states using a three-copy fusion-sector observables construction (their Eq. (9)), reporting $|S| \approx 2.633$ on $|\tau, \tau; 1\rangle^{\otimes 3}$. Their setup differs from ours in both the ambient fusion space and the observable choice; a detailed setup-by-setup comparison appears in the companion paper [6].

Recent methodological work on braid-circuit verification by Amaral [23] and on minimal-depth Levin–Wen circuits by Bseiso *et al.* [24] provides additional methodological context for the F-symbol conventions and Yang–Baxter checks used here. A companion Leggett–Garg inequality analysis on the same Fibonacci representation, saturating the Lüders bound $K_3 \rightarrow 3/2$, is reported separately in Ref. [25].

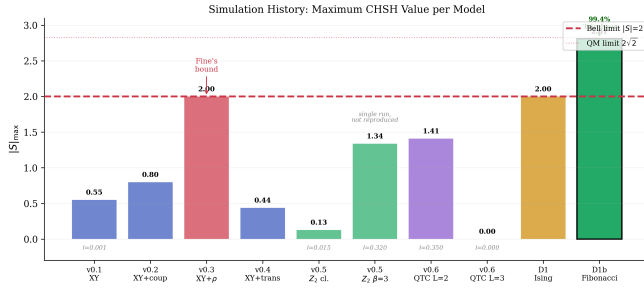


FIG. 4. Maximum CHSH value across all simulation models. The classical limit $|S| = 2$ (dashed red) is never exceeded in abelian models. Fibonacci anyons at braiding length 12 achieve $|S| = 2.811$ (99.4% of the Tsirelson bound $2\sqrt{2}$; corrected value, see Note on optimization, Sec. III G).

F. Relation to Companion Work and Open Questions

Two of the directions previously listed here have in fact already been carried out and are reported elsewhere. The convergence of $|S|$ toward the Tsirelson bound and the complete braid landscape for lengths beyond $L = 12$ are analyzed in the companion paper Ref. [6]; the temporal-correlation counterpart on the same Fibonacci representation, using Leggett–Garg-style protocols, is reported in Ref. [25]. The finite-size scaling of the $\mathbb{Z}_2$ sector–CHSH mutual information is settled within the present paper (Sec. III B), where it is shown to decay toward zero with system size. Two questions remain genuinely open. Whether a microscopic Hamiltonian realization — for example the Kitaev honeycomb model in its non-Abelian phase [15] — reproduces the idealized-TQFT sector dependence is not established here. And whether the sector-dependence generalizes to other non-Abelian anyon models, such as $SU(2)_k$ theories at level $k \geq 3$, is likewise not addressed by the present construction.

V. CONCLUSION

We have presented a systematic numerical investigation demonstrating that topological structure produces sector-dependent CHSH violation in a non-Abelian anyon model. The progression from classical $\mathbb{Z}_2$ gauge theory ($I(T; S) > 0$, 9/10 seeds $> 2\sigma$) through abelian quantum anyons (locally invisible at $L \geq 3$) to Fibonacci anyons ($|S| = 2.811$, sector-dependent) traces a clear path: only non-Abelian topological order with universal braiding generates CHSH violation through topology in the models tested here (Fig. 4).

The key finding is consistent with the topological sector neither creating nor destroying entanglement, but rotating the optimal measurement basis in fusion space — scoped to the CHSH-violation ceiling under adapted settings, since the entanglement itself (concurrence) varies substantially with sector, from 0 to 0.99 over the Fi-

bonacci δ -sweep (Sec. III G). An observer who does not know the sector sees variable $|S|$, ranging from near zero to 2.72 with fixed settings. This mechanism — topology as a structured classical degree of freedom that modulates which correlations are visible — provides a concrete analytical framework connecting topological order to the structure of quantum correlations. The $\mathbb{Z}_2$ mutual information provides a finite-size classical precursor, the Ising analysis identifies the fermion parity obstruction, and the Fibonacci result indicates that this obstruction is not universal. Together, they constitute a systematic investigation of $I(\text{topology}; \text{CHSH})$ across abelian and non-Abelian models.

DATA AND CODE AVAILABILITY

The deterministic analysis scripts and JSON output files deposited with this record reproduce the crossover decomposition of Table II, the gauge-invariant confirmation of Table III, the finite-size scaling of Tables IV and V (per-cell data in `data/stufe2_analyse.json`), the frozen-regime characterization, and Figs. 1(a) and 2; the systematic seed sets are $\{42, 10-13\}$ for the decomposition, $\{42, 10-29\}$ for the gauge-invariant recompute, and 10–19 for the scaling cells. The per-seed entries of Table I are from the original analysis and have no deposited generating script; its $L = 16$ mean is independently reproduced by the deposited scaling run (Table IV, v05 column, to within 0.03σ). The Fibonacci values of Table VI are reproduced from the companion record cited there, whose deposit recomputes them for the published braid words; the Ising saturation $S = 2.000$ follows analytically from the fermion parity superselection rule and is not a numerical result. The remaining entries of Table VII and the corresponding bars of Fig. 4 record earlier model iterations and are kept as history. All files are archived on Zenodo under the concept DOI [10.5281/zenodo.19600752](https://doi.org/10.5281/zenodo.19600752) (which always resolves to the latest version). Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the MIT license (see LICENSE and LICENSE-CODE).

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends

on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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Appendix A: Simulation Parameters and Negative Results

All classical simulations use $L = 16$ with Metropolis MC (single-site for XY, single-edge for $\mathbb{Z}_2$). The deposited engine and validation scripts thermalize for 2000 sweeps and take 4000 measurements per run (the frozen-regime characterization: 40 000 measurements); an exploratory β scan included in the engine uses 1000/2000. MI permutation test: 500–2000 shuffles. Detectors at (4, 4) and (12, 12). Quantum toric code: exact diagonalization via projector method.

TABLE VII. Complete simulation history. The $\mathbb{Z}_2$ $\beta=3.0$ row is a single early frozen-regime run, kept as history; it is not reproduced by the deposited systematic seed set (Table II).

Model	$ S _{\max}$	I (bits)	σ	Key finding
XY 16×16 ($\beta=1.0$)	0.55	0.001*	1.5*	Signal not robust (4/10 seeds)
XY + meas. coupling	0.80	–	–	Coupling adds noise
XY + $\rho(n a, b)$ weighting	2.000	–	–	Fine bound reached exactly
XY + transport phase	0.44	–	–	$\cos \theta$ form, 17% amplitude
$\mathbb{Z}_2$ gauge ($\beta=2.0$)	0.13	0.015	10.8	Robust: 9/10 seeds $>2\sigma$
$\mathbb{Z}_2$ gauge-inv. ($\beta=2.0$)	–	0.018	14.2	Confirmed: 21/21 $>3\sigma$
$\mathbb{Z}_2$ gauge ($\beta=3.0$)	1.34	0.320	590	Frozen regime (trivial)
Quantum TC $L=2$	1.41	0.350	1595	Finite-size artifact
Quantum TC $L=3$	0.00	0.000	0	Abelian: locally invisible
Perturbed TC (h -scan)	0.00	–	–	Perturbation destroys correlations
Ising anyons (Majorana)	2.000	–	–	Fermion parity: $S \leq 2$ locally
Fibonacci anyons (len=12)	2.811	–	–	CHSH violation: 99.4% Tsirelson[†]

^a * Not robust across independent seeds. [†] Updated from full enumeration; see Note on optimization, Sec. III G. Bold: primary results.

TABLE VIII. Negative results.

Test	Result	Implication
Wilson-loop CHSH ($L=3$)	All correlations zero	Stabilizers + abelian = trivial
Fourier mode $\propto n$	Dominant $k=1$ for all $ n $	No frequency shift with sector
Resonance at $\theta=\pi/n$	Ratio 1.2–1.4 (weak)	Suggestive but not significant
XY seed variation (10 runs)	4/10 $>2\sigma$	XY not robust for claim
Perturbed TC (all h)	$ S =0$ for $h>0$	Perturbation destroys $L=2$ corr.
Ising braid-then-fuse	$S=2.000$ exactly	Fermion parity superselection

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